



NORTH SYDNEY BOYS HIGH SCHOOL

2015 HSC ASSESSMENT TASK 4

Mathematics

General Instructions

- Working time – 55 minutes
- Write on the lined paper in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.
- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Ms Lee
- ☐ Mr Berry
- ☐ Ms Ziazaris
- ☐ Mr Zuber

Student Number: _____

(To be used by the exam markers only.)

Question No	1-4	5	6	7	8	Total	Total %
Mark	$\frac{\quad}{4}$	$\frac{\quad}{7}$	$\frac{\quad}{14}$	$\frac{\quad}{11}$	$\frac{\quad}{9}$	$\frac{\quad}{45}$	$\frac{\quad}{100}$

SECTION A - MULTIPLE CHOICE (4 Marks)

Question 1

Which of the following is the sum of the first n terms of the series:

$$(x + y) + (x + 2y) + (x + 3y) \dots$$

(a) $x + ny$

(b) $n(x + ny)$

(c) $\frac{(x+y)(y^{n-1}-1)}{y-1}$

(d) $\frac{n}{2}(2x + ny + y)$

Question 2

Consider the following sentence:

"House prices in Sydney are expected to rise over the next year however the they seem to be slowing and it is expected that prices will soon stabilise."

If P is the average Sydney house price and t is time, which of the following is true?

(a) $\frac{dP}{dt} > 0$ & $\frac{d^2P}{dt^2} > 0$

(b) $\frac{dP}{dt} > 0$ & $\frac{d^2P}{dt^2} < 0$

(c) $\frac{dP}{dt} < 0$ & $\frac{d^2P}{dt^2} > 0$

(d) $\frac{dP}{dt} < 0$ & $\frac{d^2P}{dt^2} < 0$

Question 3

If Timmy invests \$ 1200 on January 1 2008 into an account that pays 6% interest compounded monthly, when will he have \$2000?

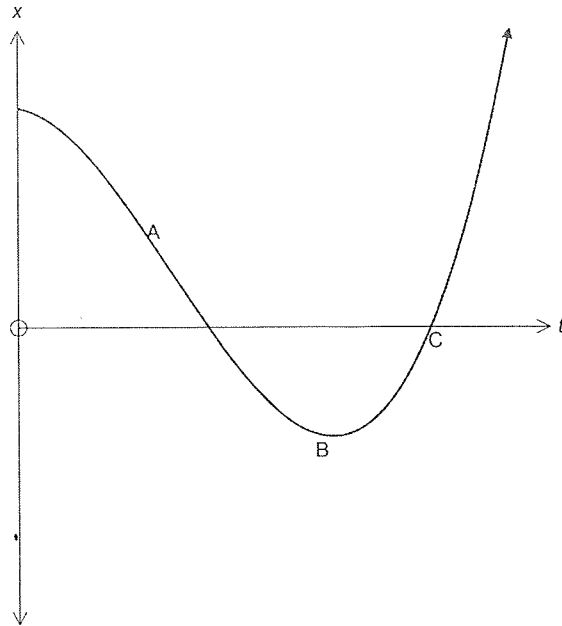
(a) July 2016

(b) May 2017

(c) April 2018

(d) February 2019

Question 4



The graph above is a displacement/time graph for a particle travelling in one dimension. At what point is the acceleration equal to zero?

- (a) A
- (b) B
- (c) C
- (d) This particle will never have an acceleration of zero.

SECTION B – EXTENDED RESPONSE

Question 5 – Sequences and Series (7 Marks)

Marks

- (a) The third and seventh term of a geometric series are $1\frac{1}{4}$ and 20 respectively,
Find:

(i) The common ratio

2

(ii) The first term

1

(iii) The 15th term

1

- (b) Evaluate:

3

$$\sum_{n=1}^{14} 1 - 2n + 2^n$$

Question 6 – Rates of Change (14 Marks)

- (a) The amount of water in a puddle over a four hour period is given by:

$$W = 4t^3 - t^4 + 20 \quad \text{where } 0 \leq t \leq 4$$

where W is the amount of water in litres and t is the time in hours.

- | | | |
|-------|---|---|
| (i) | How much water was initially in the puddle? | 1 |
| (ii) | What was the maximum amount of water in the puddle during this time? | 3 |
| (iii) | When was the amount of water increasing the most rapidly? | 2 |
| (iv) | Neatly sketch $W = 4t^3 - t^4 + 20$ for $0 \leq t \leq 4$, showing all relevant features | 2 |
- (b) The percentage of Carbon 14 in an organism falls exponentially after the death of the organism. After 1845 years 80% of the original concentration of Carbon 14 remains.
- Using the equation $C = C_0 e^{-kt}$ to represent the exponential fall of Carbon 14:
- | | | |
|-------|--|---|
| (i) | Find the exact value of k | 2 |
| (ii) | An artefact containing this organism has 65% of the original concentration of Carbon 14. How long has this organism been dead? Give your answer to the nearest year. | 2 |
| (iii) | A sea sponge containing this organism has been dead for 12000 years. What percentage (to 1 decimal place) of the original Carbon 14 concentration does it have? | 2 |

Question 7 - Applications of Series (11 Marks)

- (a) James received 3 tonnes of topsoil for his yard. He uses a wheelbarrow which can hold 150 kg to spread the soil.
- (i) How many loads in the wheelbarrow will he need? 1
- He begins at the pile of topsoil and deposits the first load 3 metres from the pile. Each successive load is dumped half a metre further from the pile.
- (ii) How far from the pile will he leave the final barrow load? 2
- (iii) What is the total distance that James will travel with the wheelbarrow if he finishes back at his starting point? 2
- (b) Pepe is saving for a new gaming computer. He opens an "Incentive Saver Account" which pays interest at the rate of 4.8% p.a. compounded monthly at the end of each month.
- Pepe decides to deposit \$40 into the account on the first of each month. He makes his first deposit on the 1st January 2015 and his last on the 1st July 2017.
- He withdraws the entire amount, plus interest, immediately after his final interest payment on 31st July 2017.
- (i) How much did Pepe deposit into his "Incentive Saver Account"? 1
- (ii) How much did Pepe withdraw from his account on the 31st July 2017? 3
(Show all relevant working)
- (iii) Pepe spends all of the money on his new computer. Five years later it is worth half of its original price. Calculate its annual rate of depreciation. 2

Question 8 – Motion (9 Marks)

- (a) A particle P is moving in a straight line so that its velocity v metres per second after t seconds is given by $v = 12 - 4t$. Initially P is 3 metres to the right of the origin O .
- (i) Find the initial velocity and acceleration of P . 2
- (ii) If the displacement of P from O is x metres, find an expression for x in terms of t . 2
- (iii) Find when and where P is stationary 2
- (iv) Sketch the graph of $v = 12 - 4t$, for $0 \leq t \leq 5$ 1
- (v) Hence, or otherwise, find the total distance travelled by P during the first five seconds. 2

End of Examination

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Section A

1. / D

2. / B

3. / A

4. / A

Question 5

Marks

(a) $T_3 = 5/4$
 $T_7 = 20$

(i) $\frac{T_7}{T_3} = \frac{ar^6}{ar^2}$

$\therefore r^4 = \frac{20}{(5/4)}$

$\therefore r^4 = 16$

$r = \pm 2$

(ii) $20 = ar^6$

$a = \frac{20}{(\pm 2)^6}$

$= 5/16$

(iii) $T_{15} = (5/16)(\pm 2)^{14}$
 $= 5120$

(b) $\sum_{n=1}^{14} (1-2n) + (2^n)$

$= \frac{14}{2} ((-1) + (-27)) + \frac{2(2^{14} - 1)}{2 - 1}$

$= -196 + 32766$

$= 32570$

Question 6

Marks

(a) (i) 20 L

✓

$$(ii) W = 4t^3 - t^4 + 20$$

$$\frac{dW}{dt} = 12t^2 - 4t^3$$

$$\text{when } \frac{dW}{dt} = 0$$

$$4t^2(3-t) = 0$$

∴ test $t=0$ & $t=3$

✓

t	-1	0	1	3	4
$\frac{dW}{dt}$	16	0	8	0	-64
W	/	-	/	-	\

✓

∴ maxima occurs when $t=3$

$$W = 4 \times 3^3 - 3^4 + 20$$

$$= 47 \text{ L}$$

✓

(iii) to maximise $\frac{dW}{dt}$, $\frac{d^2W}{dt^2} = 0$

$$\frac{d^2W}{dt^2} = 24t - 12t^2$$

$$\therefore 12t(2-t) = 0$$

$$\therefore t = 2 \text{ or } t = 0$$

✓

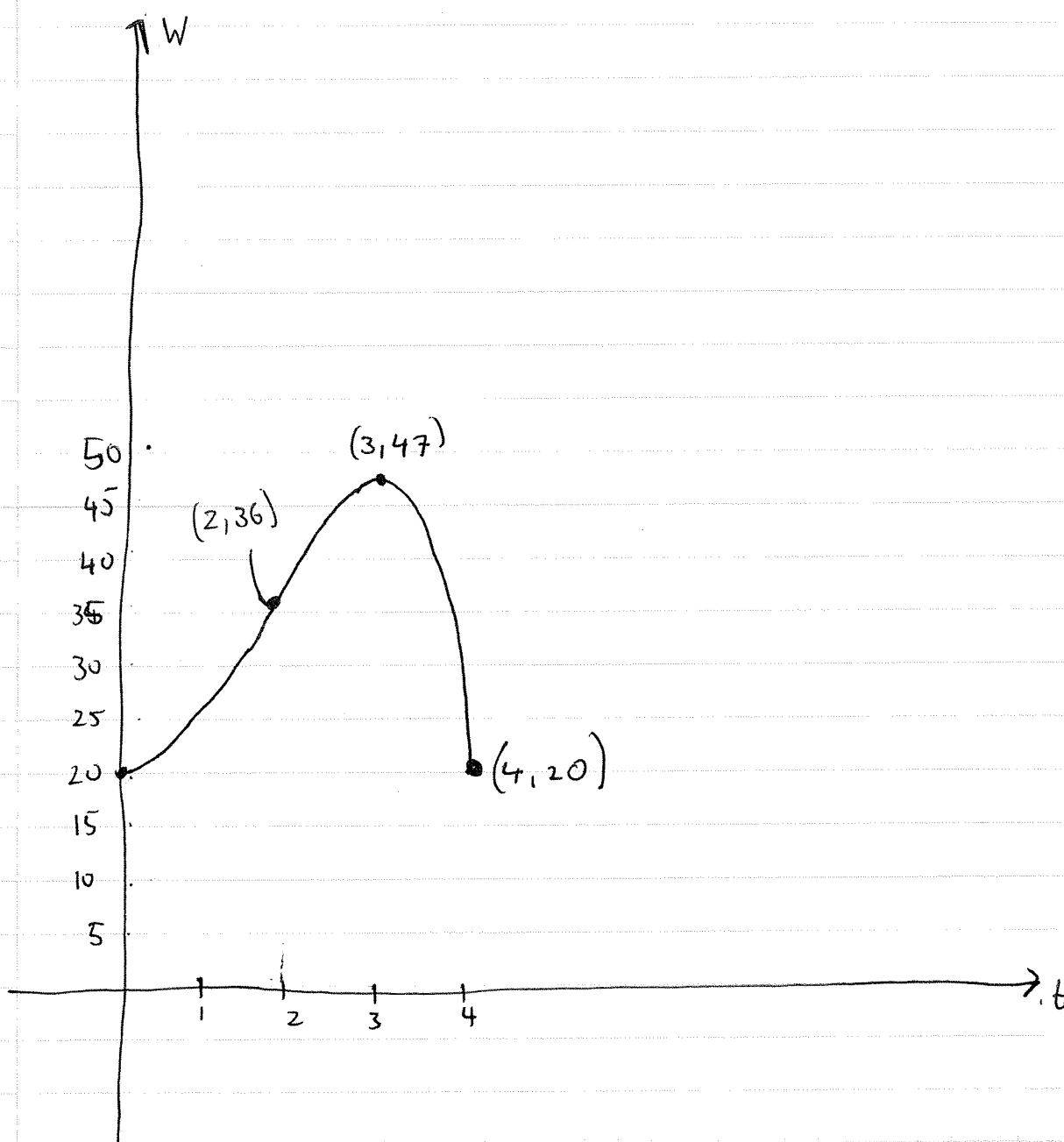
disregard $t=0$ as part (ii) established when $t=0$ $\frac{dW}{dt} = 0$

t	1	2	3
$\frac{d^2W}{dt^2}$	12	0	-36
W	4	1	0

✓

∴ P.O.I occurs at $t=2$

so most rapid increase at $t=2$



✓✓

(b) (i)

$$0.80 = e^{-1845k}$$

$$\log_e(0.80) = -1845k$$

$$\therefore k = -\frac{\log_e(0.80)}{1845}$$

$$(ii) \quad 0.65 = e^{-kt}$$

$$-kt = \log_e(0.65)$$

$$t = \frac{\log_e(0.65)}{-k}$$

$$= 3562 \text{ years (nearest yr)}$$

$$(iii) \quad A = e^{-12000k}$$

$$= 0.234256 \dots$$

$$\therefore 23.4\%$$

Question 7

Marks

$$(a)(i) \frac{3000}{150} = 20$$

✓

(ii) This is an AP where:

$$a = 3$$

$$d = 0.5$$

$$n = 20$$

✓

$$T_{20} = 3 + (20-1) \times 0.5$$

$$= 12.5 \text{ m}$$

✓

(iii) This is an AP where

$$a = 6$$

$$d = 1$$

$$n = 20$$

✓

$$S_{20} = \frac{20}{2} (2 \times 6 + (20-1) \times 1)$$

$$= 310 \text{ m.}$$

✓

$$(b)(i) 31 \times \$40 = \$1240$$

✓

$$(ii) A_{31} = 40 (1.004)^{31}$$

$$A_{30} = 40 (1.004)^{30}$$

$$A_{29} = 40 (1.004)^{29}$$

and so on until

$$A_1 = 40 (1.004)^1$$

✓

$$A = A_1 + A_2 + A_3 + \dots + A_{31}$$

$$= 40 (1.004 + 1.004^2 + 1.004^3 + \dots + 1.004^{31})$$

$$\text{since } \frac{1.004^2}{1.004} = \frac{1.004^3}{1.004^2}$$

$$= 1.004$$

$\therefore$ this is a GP where $r = 1.004$

$$\therefore A = \frac{40 \times 1.004 (1.004^{31} - 1)}{1.004 - 1}$$

$$= \$1322.63 \text{ (2 d.p.)}$$

$$\text{(iii) } 0.5 = (1-r)^5$$

$$\therefore 1-r = \sqrt[5]{0.5}$$

$$\therefore r = 0.12944$$

$$r = 13\% \text{ (nearest \%)}$$

Question 8.

Marks.

(a) (i) when $t=0$

$$v = 12 \text{ m/s}$$

$$a = -4 \text{ m/s}^2$$

(ii) $x = 12t - 2t^2 + c$

when $t=0$ $x = 3$

$$\therefore 3 = 12 \times 0 - 2 \times 0 + c$$

$$c = 3$$

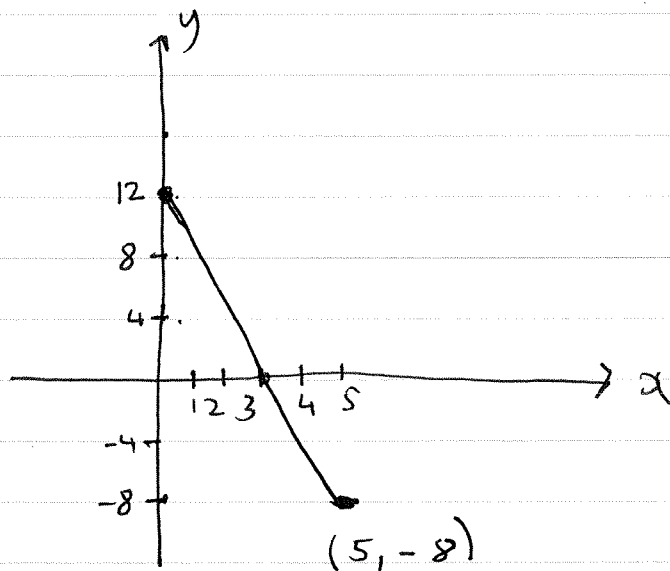
$$\therefore x = (12t - 2t^2 + 3) \text{ m}$$

(iii) when $v = 0$

$$12 - 4t = 0$$

$$t = 3, \quad x = 21 \text{ m}$$

(iv).



(v) $d = \int_0^3 (12 - 4t) dt + \left| \int_3^5 (12 - 4t) dt \right|$

$$= \left(\frac{1}{2} \times 3 \times 12 \right) + \left(\frac{1}{2} \times 2 \times 8 \right)$$

$$= 26 \text{ m.}$$